Awwab Usama Abdus Samad

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Summary

Computer Engineering graduate from ITU and Master's student in Artificial Intelligence at UT Austin with hands-on experience in machine learning and deep learning. Interned in ML crop-classification research and completed a Master's course project in infrastructure damage detection using two state-of-the-art detection models. Developed AI-driven applications, including diet prediction systems and intelligent customer chatbots. Passionate about solving complex problems through AI engineering. Actively seeking opportunities in AI research, development, or related fields.

Education

Master's of Science in Artificial Intelligence **2025 – Present**
University of Texas, Austin, USA - (CGPA: 3.93/4.00)

Bachelor's of Science in Computer Engineering **2021 – 2025**
Information Technology University, Lahore, Pakistan - (CGPA: 3.82/4.00)

Professional Experience

Machine Learning Intern **May 2024 – July 2024**
Punjab Information Technology Board (PITB) *Lahore, Punjab*

- Trained random forest and SVM models on large-scale crop datasets and Sentinel-2 satellite imagery to perform segmentation and masking, and classify crops across the Punjab province on Google Earth Engine, achieved a 92% classification accuracy.

Skills Used: Python, Google Earth Engine, VS Code, Scikit-Learn, Decision Trees, Random Forests

Undergraduate DBMS Teaching Assistant **Jan 2025 – Jun 2025**
Information Technology University (ITU) *Lahore, Punjab*

- Assisted the Database Management Systems course instructor in teaching 52 UG Software Engineering (SE23) students.
- Strengthened discipline and communication skills by organizing weekly TA sessions, managing grades, and providing support for students in SQL and database design.

Projects

Comparing YOLOv8 & RT-DETR for Road Damage Detection (Course Project) **Apr 2026**
University of Texas at Austin *Report Link*

- Trained two state-of-the-art detection models, YOLOv8 (CNN) and RT-DETR (Hybrid CNN-Transformer) on 4 categories of road distress, compared their performance and CNN/Transformer architectures, trained on Colab T4 GPU.
- Reached best mean absolute precision (mAP) of 0.548 for RT-DETR (vs. 0.522 of YOLOv8) and a 60% relative performance boost in detecting rare potholes, YOLOv8 trained 3x faster with a 4ms faster inference speed.
- Authored a technical paper in APA format on Latex to document the research, methodology, results, and tradeoffs of the architectures.

Skills Used: Python, Jupyter Notebook, Google Colab, Overleaf, Ultralytics Library

RAG-Based Assistant Chatbot for an E-Commerce Website (Final Year Project) **Feb 2025**
Information Technology University *GitHub Link*

- Developed an AI-powered customer support chatbot for an e-commerce marketplace for our final year project using OpenAI GPT-3.5 to generate real-time, product-aware responses with a latency of <2 seconds.
- Designed a MongoDB vector-search pipeline with weighted indexes to retrieve and rank relevant products, and then enrich the chatbot responses with pricing, discounts, and ratings, integrated via a FastAPI backend and Next.js frontend.

Skills Used: Python, FastAPI, OpenAI API, TypeScript, TailwindCSS, MongoDB

Courses & Certifications

- **AWS AI Practitioner Challenge (AWS AI & ML Scholars)** — AWS **Completed: Jun 2026**
  Skills Gained: Gen AI/LLM Architectures, AWS SageMaker & Bedrock, MLOps, RAG Systems, Prompt Engineering
- **AWS Academy Cloud Foundations** — AWS **Completed: Dec 2024**
  Skills Gained: VPC Networking, IAM Access, EC2 Instances, S3 Buckets & RDS Databases

Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScrpt, C/C++
- **ML Libraries & Frameworks:** TensorFlow, PyTorch, Scikit-Learn, OpenCV, NumPy, Pandas Matplotlib
- **Deep Learning:** Neural Networks, CNNs, Transformers, Object Detection
- **Developer Tools:** Git, VS Code, Jupyter Notebook, Google Colab, AWS